

Multiple Imputation for Kernel Density Estimation

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Abstract

The text of your abstract. 200 or fewer words.

Keywords: 3 to 6 keywords, that do not appear in the title

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1 Introduction

Assessing the distribution of an observed variable is an issue of keen interest to many areas of statistical practice. Density estimates are often part of exploratory data analysis (EDA) to help an analyst understand how the observed data are distributed. In addition, many statistical procedures rely on specific distributional assumptions about the data: classic parametric tests such as a t-test rely on the assumption that the data come from a normal population; Traditional linear regression analysis assumes the model residuals follow a normal distribution; data that follows a multimodal distribution may lead a practitioner to perform mixture modeling.

When data are fully observed there are a host of methods for estimating the probability density function; here we focus on kernel density estimation.

Elmore et al. (2006) Dubnicka (2009) Based on Horvitz-Thompson estimator (Horvitz & Thompson (1952))

Saloui: <https://www.tandfonline.com/doi/abs/10.1080/03610926.2016.1205618>

2 Sim settings

Sample size? Number of imputations? Imputation method: pmm, norm, etc?

True Distribution of Y: 1. Normal 2. Chi squared 3. Mixture of Normals

Missingness: 1. MCAR 2. MAR 3. MNAR?????

Missing rate: 1. Low 15ish 2. Medium 30ish 3. High 50ish

Combining Rules: 1. stacking and then one density estimate 2. estimating and the combining

3 Describe the combining rules

Methods: - Stack

Suppose we have B imputed data sets and we estimate f using a standard kernel density estimator given by

$$\hat{f}_b(x) = \frac{1}{nh_b} K\left(\frac{x - x_i}{h_b}\right) \quad b = 1, 2, \dots, B, \quad (1)$$

where K is an appropriately chosen kernel function and h_b is a bandwidth select, see [Simonoff \(2012\)](#) for more details. We propose to estimate f by averaging over the individual B estimators, that is

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b. \quad (2)$$

Clearly, $\hat{f}(x)$ is a proper density in that it integrates to one and is non-negative everywhere.

4 Methods

5 Real data

Fish: Aging the fish you have to kill the fish.

AIDS:

6 Conclusion

7 Disclosure statement

The authors have the following conflicts of interest to declare (or replace with a statement that no conflicts of interest exist).

8 Data Availability Statement

Deidentified data have been made available at the following URL: XX.

SUPPLEMENTARY MATERIAL

9 BibTeX

We encourage you to use BibTeX. If you have, please feel free to use the package natbib with any bibliography style you're comfortable with. The .bst file agsm has been included here for your convenience.

References

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Horvitz, D. & Thompson, D. (1952), ‘A generalization of sampling without replacement from a finite universe’, *Journal of the American Statistical Association* **47**, 663–685.

Simonoff, J. S. (2012), *Smoothing methods in statistics*, Springer Science & Business Media.